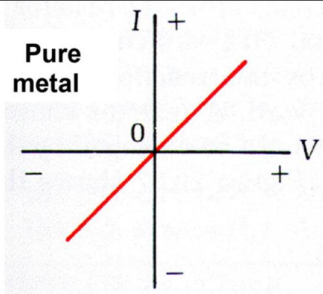
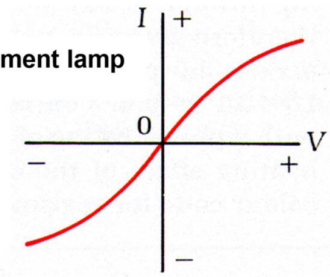
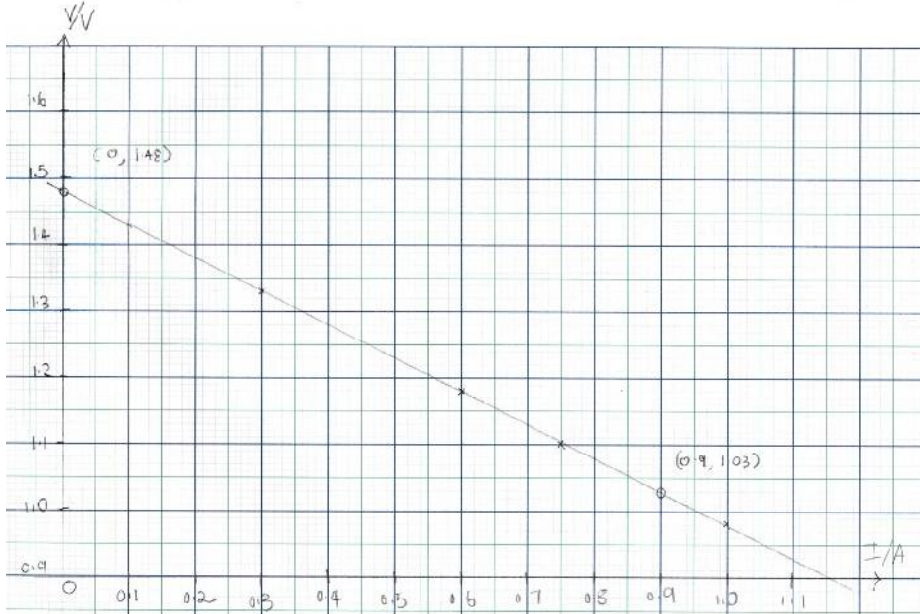


7 (a)(i)	Resistance is the ratio of the potential difference across the component to the current passing through it	[1]
(ii)	Pure metal  Filament lamp 	[2]
(b)	$R = \frac{\rho l}{A} = \frac{1.7 \times 10^{-8} \times 1.5}{3.2 \times 10^{-9}}$ $= 8.0 \, \Omega$	[1] [1]
(c)	$E = IR$ $12 = I \times (1.0 + 2.0 + 8.0)$ $I = 1.091 \, \text{A}$ $I = nAqv$ $1.091 = (8.5 \times 10^{28}) (3.2 \times 10^{-9}) (1.6 \times 10^{-19}) v$ $v = 0.025 \, \text{m s}^{-1}$	[1] [1]
(d)(i)	Using potential divider equation, $V_{PQ} = \frac{R_{PQ}}{R_{total}} \times 12.0 = \frac{8.0}{8.0 + 1.0 + 2.0} \times 12.0$ $= 8.7 \, \text{V}$	[1]
(ii)	terminal pd of B = $\frac{R}{R+r} \times E = \frac{4.0}{4.0 + 3.0} \times 1.5$ $= 0.857 \, \text{V}$ $\frac{V_{PJ}}{V_{PQ}} = \frac{I_{PJ}}{I_{PQ}}$	[1]

	Since terminal pd of B = V_{PJ} $\frac{0.857}{8.7} = \frac{I_{PJ}}{1.5}$ $I_{PJ} = 0.148 \text{ m}$	[1]
		[1]
(iii)	To improve accuracy, the modification is either to: <u>replace cell A with one of lower emf</u> or reducing the resistance of the wire through replacing the copper wire with one of <u>higher cross-sectional area</u>, <u>or lower resistivity</u> or <u>replace the 2.0Ω resistor with one of higher resistance</u> or <u>add in another resistor in series with cell A.</u> Explanation: To <u>reduce the fractional/ percentage error</u> in the determination of balance length. (do not accept: "longer balance length" alone)	[1]
(e)(i)	 Best-fit drawn shown Since, $V = E - Ir$, $V = -rI + E$ y-intercept = $E = \underline{1.48 \text{ V}}$ gradient = $\frac{1.48 - 1.03}{0.00 - 0.90} = -0.50$ $\Rightarrow r = \underline{0.50 \Omega}$	[1]
		[1]
		[1]
		[1]
(ii)	As temperature increases, <u>resistance of thermistor decreases</u>. The effective resistance of the circuit decreases. <u>Total current increases</u>, hence from $V = E - Ir$, <u>terminal potential difference/ p.d. across R decreases</u>.	[1]
		[1]

	Since $P = \frac{V^2}{R}$, power dissipated in resistor R <u>decreases</u>. (Alternatively, As T increases, <u>$R_{\text{thermistor}}$ decreases</u>, <u>Effective resistance across R & thermistor decreases</u>, (by potential divider), <u>$V_{\text{thermistor}}$ takes a smaller share/proportion of the supply p.d.</u>, thus by $P = \frac{V^2}{R}$, <u>P decreases</u>)	[1] [1] [1] [1]
--	--	---